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ASSESSMENT TOOL – 6

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COURSE NAME: Cloud Computing and Big Data Analysis

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

ANSWER:

Introduction

A media company wants to migrate its video streaming platform to the cloud. The platform must support millions of users with minimal buffering and high availability.

Cloud Architecture

- Users access the streaming application through the Internet.
- A **Content Delivery Network (CDN)** caches videos close to users.
- A **Load Balancer** distributes incoming requests across multiple application servers.
- **Auto-Scaling** automatically adds or removes servers based on demand.
- Videos are stored in **cloud object storage**.
- A cloud database stores user accounts, subscriptions, and watch history.

Load Balancing

- Distributes traffic evenly among servers.
- Prevents server overload.

- Improves availability and performance.

Auto-Scaling

- Automatically launches additional servers during peak traffic.
- Removes unused servers during low traffic.
- Maintains smooth video streaming.

Caching Techniques

- Store frequently watched videos in CDN edge servers.
- Cache application data using in-memory caching.
- Reduce requests to the main storage.

Cost Optimization

- Use auto-scaling to pay only for required resources.
- Store old videos in lower-cost archive storage.
- Compress videos to reduce storage and bandwidth.
- Use reserved instances for predictable workloads.

Conclusion

Using CDN, caching, load balancing, auto-scaling, and cloud storage provides high performance, reduced buffering, and cost-efficient streaming.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

ANSWER:

Introduction

A FinTech startup deploys applications across multiple cloud providers using containers.

Proposed Solution

- Package applications as Docker containers.
- Use Kubernetes to manage containers across multiple clouds.
- Deploy clusters in different cloud providers for high availability.

Service Discovery

- Kubernetes automatically assigns service names.
- Containers communicate without knowing IP addresses.
- Internal DNS enables easy service discovery.

Scaling

- Kubernetes Horizontal Pod Autoscaler increases or decreases container instances based on CPU or memory usage.
- Load balancers distribute traffic among container replicas.

Benefits of Multi-Cloud

- High availability.
- Disaster recovery.
- Avoids vendor lock-in.
- Better global performance.

Risks

- More complex management.
- Higher networking costs.
- Security policy differences between providers.
- Increased operational complexity.

Conclusion

Container orchestration with Kubernetes provides consistent deployment, automated scaling, and simplified management across multiple cloud providers.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

ANSWER:

Introduction

An e-learning platform loses data due to accidental deletion and requires a strong backup and disaster recovery solution.

Proposed Solution

- Schedule automatic cloud backups.
- Replicate data to another cloud region.
- Enable object versioning to recover deleted files.
- Use snapshots for databases and virtual machines.

Recovery Time Objective (RTO)

- Target: Restore services quickly.
- Achieved by keeping replicated systems ready for failover.
- Automated recovery minimizes downtime.

Recovery Point Objective (RPO)

- Target: Minimize data loss.
- Achieved through frequent backups and continuous replication.
- Only a small amount of recent data may be lost.

Business Continuity Steps

1. Enable automatic daily backups.
2. Replicate data to another region.
3. Test disaster recovery regularly.
4. Monitor backup status continuously.
5. Perform automatic failover during failures.
6. Restore services from backup if needed.

Conclusion

Cloud-native backup, replication, and versioning provide fast recovery, minimal data loss, and continuous business operations.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

ANSWER:

Introduction

A government agency stores sensitive data on-premises while using the public cloud for non-sensitive applications.

Hybrid Cloud Architecture

- Sensitive applications remain in the on-premises data center.
- Public cloud hosts websites and non-sensitive services.
- Secure VPN or dedicated private connection links both environments.
- Data is synchronized securely between locations.

Identity Management

- Implement Single Sign-On (SSO).
- Use Identity and Access Management (IAM).
- Apply Role-Based Access Control (RBAC).
- Enable Multi-Factor Authentication (MFA).

Encryption

- Encrypt stored data (Data at Rest).
- Encrypt transmitted data using SSL/TLS (Data in Transit).
- Manage encryption keys securely.

Challenges

- Managing multiple environments.
- Increased operational complexity.
- Consistent security policies.
- Network latency.
- Compliance and governance.

Conclusion

A hybrid cloud securely combines on-premises and cloud resources using secure connectivity, IAM, encryption, and centralized management.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

ANSWER:

Introduction

A SaaS provider requires its application to remain available even if one cloud region fails.

Multi-Region Deployment Strategy

- Deploy identical applications in two or more cloud regions.
- Place a global load balancer in front of all regions.
- Route users to the nearest healthy region.
- Keep standby resources ready for failover.

Data Replication

- Replicate databases between regions.
- Use synchronous replication for critical data (where latency allows).
- Use asynchronous replication for distant regions to improve performance.
- Store backups in multiple regions.

Failover Mechanism

- Continuously monitor application health.
- If one region fails, automatically redirect traffic to another region.
- Restore failed services without user interruption.

Monitoring and Alerting

Monitor:

- CPU utilization
- Memory usage
- Disk space
- Network latency
- Application response time
- Database performance

Alerts:

- High CPU or memory usage
- Server failures
- Database replication errors
- Network outages
- Security incidents

Conclusion

A multi-region deployment with global load balancing, data replication, automatic failover, and continuous monitoring ensures high availability, fault tolerance, and uninterrupted service even during regional failures.